MIL-S-18570C(AS)
8 January 1982
Superseding
MIL-S-18507B(AS)
7 June 1976

MILITARY SPECIFICATION

SIGNAL, SMOKE AND ILLUMINATION, AIRCRAFT MARK 6 MOD 3

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the manufacture, assembly and packaging of the Signal, Smoke and Illumination, Aircraft, Mk 6 Mod 3 and the methods of examination and tests upon which product acceptance shall be based.

1.2 Definitions. The meaning of terms used in this specification shall be as listed in 6.4.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1370

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command

LD 295906	Signal, Smoke and Illumination, Aircraft, Mk 6 Mod 3
LD 261100	Signal Container, Mk 5 Mod 0
LD 261108	Shipping Box

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Item description. This prime item is the Mk 6 Mod 3 Aircraft Smoke and Illumination Signal. The signal is approximately 20.367 inches long by 5.062 inches (square). The signal weighs approximately 15 pounds (CALC). The signal consists of a wood body with a flat, die-cast metal plate affixed to one end to protect it from water impact damage and to assure proper floating attitude. The body contains four (4) pyrotechnic candles, a Mk 245 friction igniter and time fuse assembly.

3.2 Classification of requirements. The requirements for the signal are classified herein as follows:

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REQUIREMENTS	REQUIREMENT PARAGRAPH	EXAMINATION OR TEST PARAGRAPH
First Article	3.3	4.3.1
Design and Construction	3.4	4.4
Performance Characteristics	3.5	4.5.1
Workmanship	3.6	—

3.3 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.4 Design and construction

3.4.1 Prime item documentation. This signal shall be manufactured, examined, tested, packed and packaged in accordance with LD 295906 and this specification.

3.5 Performance characteristics

3.5.1 Performance

3.5.1.1 Sealing. The signal shall withstand a vacuum of 6.0 ± 1.0 inches of mercury (24 inches of mercury absolute pressure) for a minimum period of 15 seconds without signs of leakage (see 4.5.1.1).

3.5.1.2 Function. The signal shall meet the following requirements (see 4.5.1.2).

a. Function (produce visible flame and smoke).

b. Delay time 60 seconds minimum to 120 seconds maximum from time of launching to visible flame and smoke.

c. Produce flame and smoke for a minimum period of 40 minutes to a maximum period of 80 minutes.

3.6 Workmanship. The product, in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. All components shall be free of chips, dirt, grit, or other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents employed.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article (see 4.3)

b. Quality Conformance Inspections (see 4.4)

4.3 First article. The contractor shall deliver a sample of 28 signals to the destination specified (see 6.3) for examination, testing, evaluation and acceptance as specified in 4.3.1. The sample shall be prepared for delivery as specified in Section 5.

4.3.1 First article inspection. A sample of 25 signals shall be tested as specified in 4.5.1.1 and 4.5.1.2. The three remaining signals shall be used as spares. Failure of one or more signals to meet the requirements of 3.5.1.1 shall reject the sample. Failure of four or more signals to meet the requirements of 3.5.1.2 shall reject the sample.

4.4 Quality conformance inspection. An inspection lot shall consist of 501 to 3200 signals offered for delivery at one time including test samples (see 6.2.1). Inspection lots shall be inspected as follows:

a. Select for the functioning test signals from each inspection lot in accordance with MIL-STD-105, Inspection Level S-4, AQL 6.5 percent defective. Prior to functioning test, all signals of the sample shall meet the sealing requirements of 3.5.1.1. Failure of one or more signals to meet the requirements of 3.5.1.1 shall reject the inspection lot.

b. Any change in materials, suppliers or production methods shall be considered an interruption for the beginning of a new inspection lot. Record reason for interruption on the Data Card (see 6.2.2).

4.5 Tests. Random samples of signals shall be selected, tested and accepted as specified in 4.3 or 4.4, as applicable.

4.5.1 Performance tests

4.5.1.1 Sealing test. The signal, when packed in accordance with 5.1.1.1, shall be tested in a water tank equipped for observation under vacuum.

CAUTION: The water tank shall be equipped so that the signal must be raised above the water level while still under vacuum before releasing the vacuum and removing the signal.

The uppermost surface of the signal shall be submerged 2 to 6 inches below the water surface and subjected to a vacuum of 6.0 ± 1.0 inches of mercury (24 inches of mercury absolute pressure) for a minimum period of 15 seconds. Leakers are indicated by air bubbles issuing from the signal. Do not mistake the escape of occluded air for leakage. Defectives are signals failing to meet the requirements of 3.5.1.1.

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4.5.1.2 Functioning test. The signals shall be tested by launching from aircraft at an altitude of 500 feet while traveling at a relative ground speed of 150 knots into water at least 20 feet deep. The observer shall observe, record and report the following data for the test report of each lot tested (see 6.2.2):

a. Function

b. Delay time (from launching to visible flame and smoke)

c. Burning time of flame and smoke

d. Numerical sequence of signal in the test

e. Weather and water conditions at time of testing.

Defectives are signals failing to meet the requirements of 3.5.1.2.

4.6 Examinations

4.6.1 Signal Examinations. Examinations shall be made to assure that the signals meet the requirements of drawings listed on LD 295906. Only signals containing no defects shall be used for further testing or returned to the lot.

4.6.2 Packaging, packing and marking examinations. Packaging, packing and marking of the signals shall conform to Section 5 of this specification.

5. PACKAGING

5.1 Packaging and marking

5.1.1 Level A

5.1.1.1 Each signal shall be packaged in accordance with LD 261100.

5.2 Packing and marking

5.2.1 Level A

5.2.1.1 Four (4) signals packaged as in 5.1.1.1 shall be packed in accordance with LD 261108.

5.3 Ammunition data card. Data cards shall be prepared in accordance with MIL-STD-1167 (see 6.2.2).

5.4 Ammunition lot number. Ammunition lot number shall be in accordance with MIL-STD-1168.

6. NOTES

6.1 Intended use. The signal is launched from surface or aircraft to produce a day or night floating reference point.

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6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Quantity being ordered.

c. First article samples, if applicable and test facility where sample is to be shipped for evaluation.

d. Weapons Quality Engineering Center coordinating surveillance.

e. Inspection lot size, if different than 4.4.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR-7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

PARAGRAPH NO.	DATA REQUIREMENT	APPLICABLE DID	OPTIONS
4.5.1	Test/Report	DI-T-2072	---
5.3	Ammunition Data Card	DI-L-1410	---

Test reports, with individual data, will be forwarded to the Technical Activity and Design Agent.

(Data item descriptions related to this specification and identified in Section 6, will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 28 signals. The Contracting Officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the documents' first article.

6.4 Definitions

6.4.1 Signal. The term "signal" means Signal, Smoke and Illumination, Aircraft, Mk 6 Mod 3 LD 295906.

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6.5 Conflicting requirements. Refer in writing to the procuring agency for clarification, any conflicting requirements arising between this specification and any specification or publication listed herein.

6.6 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, or Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Armed Services Procurement Regulations (ASPR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY - AS

(Project 1370-N150)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-18570C(AS) Signal, Smoke and Illumination, Aircraft Mk 6 Mods 3

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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DD FORM 1426 1 OCT 76

Replaces edition of 1 Jan 72 which may be used.

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